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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B.Tech (CBCS) Degree Examinations

December 2024

IS3241 – CLOUD COMPUTING

(For IS)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. **Two fundamental and dominant models of computing are:**
A) Sequential and Parallel
B) Parallel and Distributed
C) Architecture and Compilers
D) Problem solving and Application
2. _____ **computing refers to applications and services that run on a distributed network using virtualized resources.**
A) sequential
B) Mainframe
C) cloud
D) Parallel
3. _____ **defines a service provisioning model for compute services in which resources are packaged and offered on a pay-per-use basis.**
A) Utility-Oriented computing
B) Grid Computing
C) Cluster computing
D) Mainframes
4. **Processing Elements in _____ machines work asynchronously**
A) SIMD
B) MISD
C) MIMD
D) DISD
5. **A _____ is a communication mechanism that allows the cooperation and coordination among components.**
A) Pipe
B) Filter
C) Connector
D) Program
6. **Who is responsible for deciding the system resources to be provided to the VM?**
A) Dispatcher
B) Interpreter
C) Allocator
D) Hypervisor
7. _____ **interfaces applications to libraries and/or the underlying operating systems.**
A) ISA
B) ABI
C) API
D) Interrupt
8. **A _____ is an aggregation of hosts that communicate with each other as if they were located under the same broadcasting domain**
A) VLAN
B) SAN
C) NAT
D) VPN
9. **In a community cloud, who typically shares the infrastructure?**
A) General public
B) Multiple organizations with shared concerns
C) A single organization
D) Cloud service providers
10. **What is Azure Blob Storage primarily used for?**
A) Relational database storage
B) Virtual machine backups
C) Network Security
D) Large-scale unstructured data storage

11. _____ is an XML-based language for the description of web services
 A) WSDL B) SOAP
 C) REST D) HTTP
12. Which hypervisor is also known as bare-metal hypervisor?
 A) Type 1 B) Type 2
 C) Type 3 D) Type 0
13. Address space virtualization used in time sharing systems is an example for Which of the following hardware virtualization techniques?
 A) Hardware-assisted Virtualization B) Full Virtualization
 C) Para Virtualization D) Partial Virtualization
14. What is a key benefit of using cloud-based email services like Gmail?
 A) Requirement for desktop installation B) Limited email storage
 C) High latency D) Access from any internet-connected device
15. In a market-driven cloud, the concept of elasticity refers to:
 A) The flexibility of data storage B) The ability to scale resources up or down
 C) The physical expansion of data centers D) The adjustment of network bandwidth
16. Inter Cloud architecture is designed to:
 A) Isolate cloud services from each other B) Increase the complexity of cloud management
 C) Reduce the number of cloud services D) Enable seamless integration between different cloud providers
17. In SDN, which layer is responsible for the actual forwarding of packets?
 A) Application layer B) Control layer
 C) Infrastructure layer D) Presentation layer
18. Which of the following is a challenge associated with implementing SDN?
 A) Lack of network flexibility B) Scalability issues with the SDN controller
 C) Inability to automate network management D) Increased hardware costs
19. How can you lower data center cooling costs in an energy-efficient way?
 A) Deploy fewer VMs B) Build servers that run at lower temperatures
 C) Use water to cool your data center D) Build servers that run at higher temperatures
20. Micro services is most closely related to what other established acronym?
 A) SOA B) API
 C) ABI D) ISA

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Define Cloud Computing? Explain the various risks associated with cloud computing.	8	L*2	1	2
	b) Explain with neat diagrams the Flynn's Classification models.	8	L2	1	1
2.	a) Explain distributed object programming model with a neat diagram.	8	L2	1	2
	b) Explain Remote Procedure call with neat diagram.	8	L2	1	1
3.	a) Explain with a neat diagram the virtualization reference model and discuss its use in Cloud.	8	L2	2	2
	b) Distinguish between type-1 and type-2 hypervisor.	8	L3	2	1
Unit – II					
4.	a) Explain with neat diagram the Cloud reference model architecture.	8	L2	3	2
	b) Explain the IaaS architecture with a neat diagram.	8	L2	3	1
5.	a) Briefly explain the architecture of Google AppEngine.	8	L2	3	1

b)	Describe the Core components of Windows Azure platform with a neat diagram.	8	L2	3	2
6. a)	Explain how cloud computing is applicable to Satellite image processing.	8	L2	3	2
b)	Explain the application of cloud computing in social networks and media.	8	L2	3	1

Unit – III

7. a)	Explain Market-oriented cloud computing scenario with a neat diagram.	8	L2	4	2
b)	Explain green cloud computing architecture with a neat diagram.	8	L2	4	1
8. a)	Explain the third party cloud provider MetaCDN.	8	L2	5	2
b)	With a neat diagram explain the Software Defined Network architecture.	8	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
